

RECSYS CHALLENGE 2026 · MUSIC-CRS

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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47,071

tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark gray color.



Contents

SIX PARTS

OPTIONAL

1	The problem	finding music by talking
2	Where the data comes from	how the organisers built the conversations
3	How it is scored	one right track per turn, plus a judged reply
4	Our approach	request understanding, retrieval, reranking, response generation
5	How it scored	the official results
6	What is not ideal	what went wrong, one finding per slide
A	Appendix	every other detail



1 The problem

finding music by talking





Finding music by talking

THE PROBLEM

OPTIONAL

I'm trying to remember a song that feels like a quiet, personal story...



Neko Case — "Calling Cards"

Not quite the one... it definitely had a female singer.



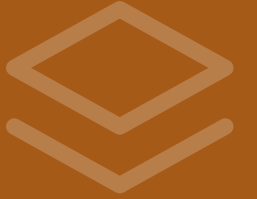
Neko Case — "Man"

No, that's not it either. Much more subdued... almost a cappella.



Your system answers here: **20 tracks** and a **short reply**

A real conversation from the challenge's final test set.



2

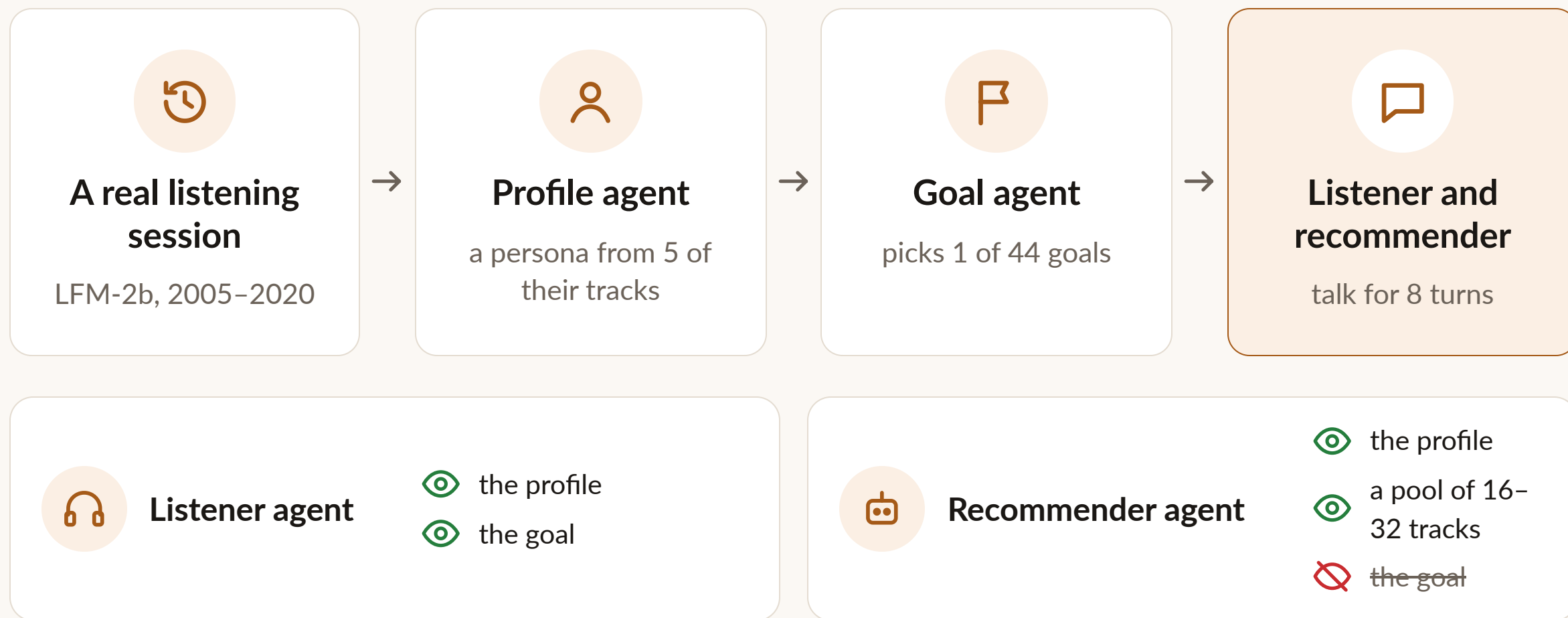
Where the data comes from

how the organisers built the conversations



How the conversations were made

THE DATA OPTIONAL



All four agents are **Gemini 2.5 Flash**. The recommender's pick **is the answer key**.

Source: TalkPlayData 2 (Doh et al., 2025), §3



What you get

THE DATA

OPTIONAL



15,199

**training
conversations**

8 turns each, with the
answer for every turn



1,000

**development
conversations**

with answers — for
checking yourself



2 × 80

**blind test
conversations**

final turn only; scored
on the leaderboard



47,071

**tracks in the
catalog**

title, artist, tags, plus
text, audio and cover
embeddings



3 How it is scored

one right track per turn, plus a judged reply





The final score

SCORING

OPTIONAL

50%

right track · nDCG@20

30%

reply quality · LLM judge, 1–5

10%

catalog
diversity

10%

reply wording

LLM JUDGE

The organisers' LLM reads the conversation and your reply, and rates it 1–5.

CATALOG DIVERSITY

How many **different** tracks you recommend across all turns, as a share of the catalog.

REPLY WORDING

Lexical diversity: how varied the words in your replies are.



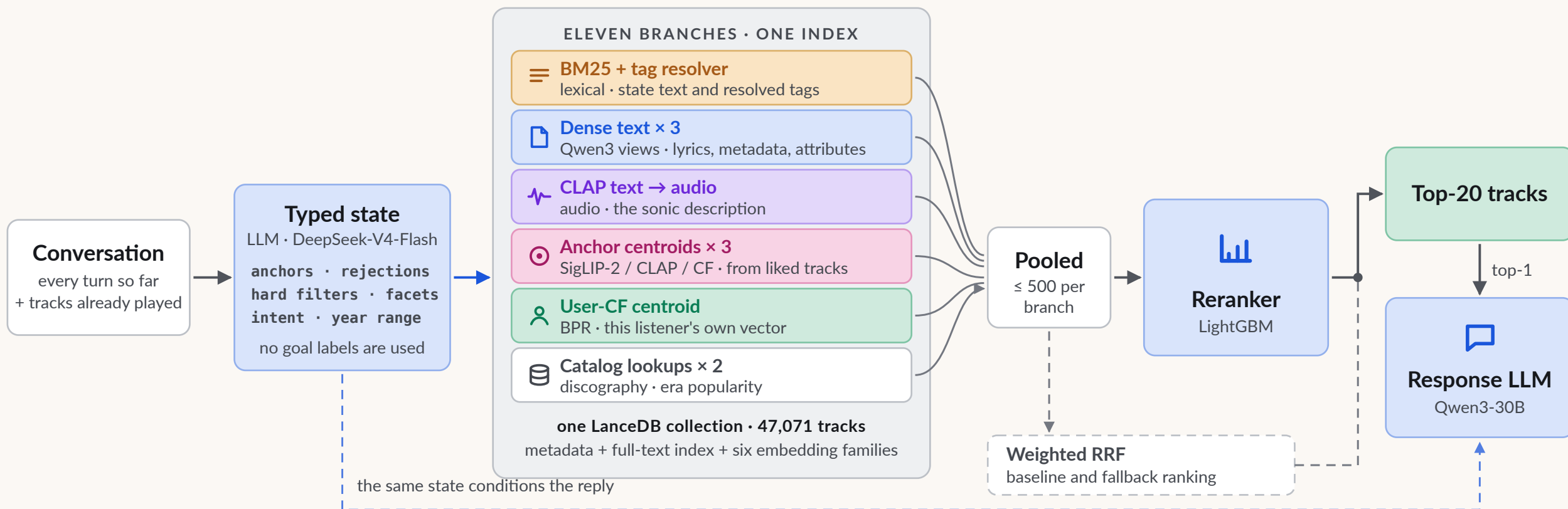
4 Our approach

request understanding, retrieval, reranking, response generation



≡ One state, eleven branches, one learned ranker

APPROACH CORE



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.



Request understanding

APPROACH **CORE**

TURN 1

🎵 “Calling Cards” — *not quite*

TURN 2

🎵 “Man” — *not it either*

TURN 3 · NOW

Much more subdued... a sense of place in the lyrics... almost a cappella.

LLM



STAY WITH

Neko Case

TURNED DOWN

“Calling Cards”, “Man”

MOOD

subdued

LYRICS

a sense of place

SOUND

stark, almost a cappella

The LLM reads **every turn so far** and writes structured fields; every later step reads those.



What the state can hold

APPROACH

OPTIONAL

THE REQUEST

- request type
- intent mode
- routing flags

WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

WHAT IT IS ABOUT

- lyrical theme

BOUNDS

- release-year window
- process constraints

Understanding a lookup

APPROACH

OPTIONAL

LISTENER

Which song from Peter Doherty's *Grace/Wastelands* was mentioned as one of the best tracks of 2009?

WE SERVED

 Peter Doherty — “Arcady”, from *Grace/Wastelands*

LLM



REQUEST TYPE

exact album

ARTIST

Peter Doherty • must use

ALBUM

Grace/Wastelands • must use

ALSO

“best tracks of 2009”

Two **hard** fields (artist, album) and one **soft** phrase. The request type switches on exact lookups.

Understanding a visual request

APPROACH

OPTIONAL

LISTENER

An album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.

LLM



REQUEST TYPE	attribute search
COVER ART	striking, abstract
COLOURS	bold colors and shapes
STYLE	like a modern painting
FLAG	visual search

No artist, no title — only **descriptions**, plus a flag that turns on the cover-art search.

Understanding “someone new”

APPROACH

OPTIONAL

LISTENER

Can you suggest more tracks with that kind of emotional depth... from other singer-songwriter artists?

LLM



REQUEST TYPE	new artist
AVOID	Ryan Adams & The Cardinals
MOOD	introspective, emotional depth
SOUND	acoustic-driven
GENRE	singer-songwriter, alt-country

“Someone new” becomes an **artist to avoid**; the mood and genre carry over.

Retrieval — eleven branches

APPROACH **CORE**



×1

BM25 + tag resolver

keywords and resolved tags
from the state



×3

Dense text

Qwen3 embeddings of lyrics,
metadata, attributes



×1

CLAP text → audio

the sonic description, matched
to audio



×3

Anchor centroids

near liked tracks: cover art,
audio, listening



×1

User-CF centroid

this listener's own BPR vector



×2

Catalog lookups

the artist's discography; era
popularity

Each branch returns its **top 500**; the ranker sees their union.



One index

APPROACH

OPTIONAL

WHAT IT DESCRIBES	MODEL	SOURCE
Lyrics	Qwen3-Embedding-0.6B	organisers
Metadata text	Qwen3-Embedding-8B	we built
Attributes text	Qwen3-Embedding-8B	we built
Audio	LAION-CLAP	organisers
Cover art	SigLIP-2	organisers
Listening data	BPR collaborative filtering	organisers
Conversation match	our fine-tuned bi-encoder	we built

ONE LANCEDB TABLE

Metadata, a full-text index and every embedding for all 47,071 tracks — each branch is a different query against the same table.

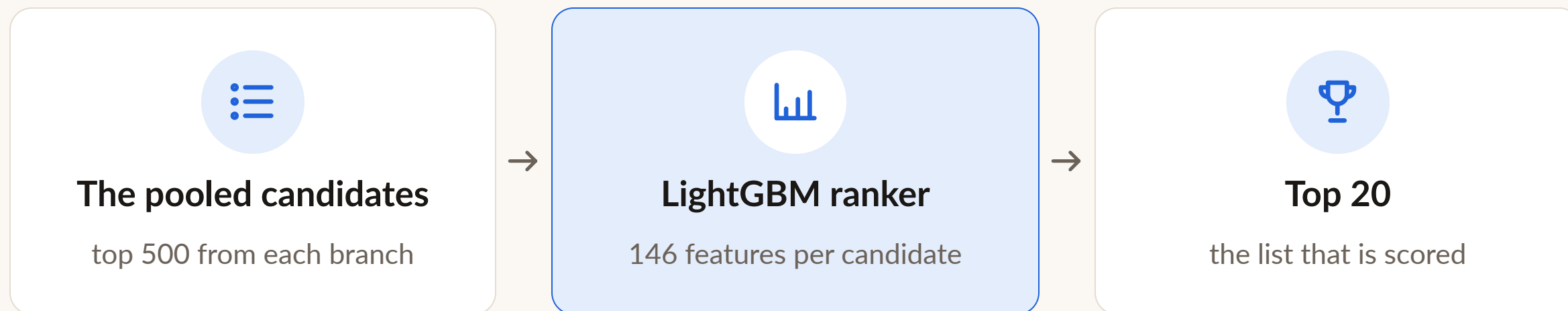
PROVIDED VS BUILT

The organisers supplied four kinds of embedding. We added larger text embeddings and our own bi-encoder.

Reranking

APPROACH

OPTIONAL





What the ranker leans on

APPROACH CORE

FEATURE FAMILY	#	GAIN
Bi-encoder cosine	1	59.0%
Per-branch rank and score	66	11.5%
Session and artist affinity	18	10.7%
Popularity · state · tags	31	9.2%
Other similarity cosines	17	8.9%
Constraints and rejections	13	0.7%

SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

GUARDS AFTER SCORING

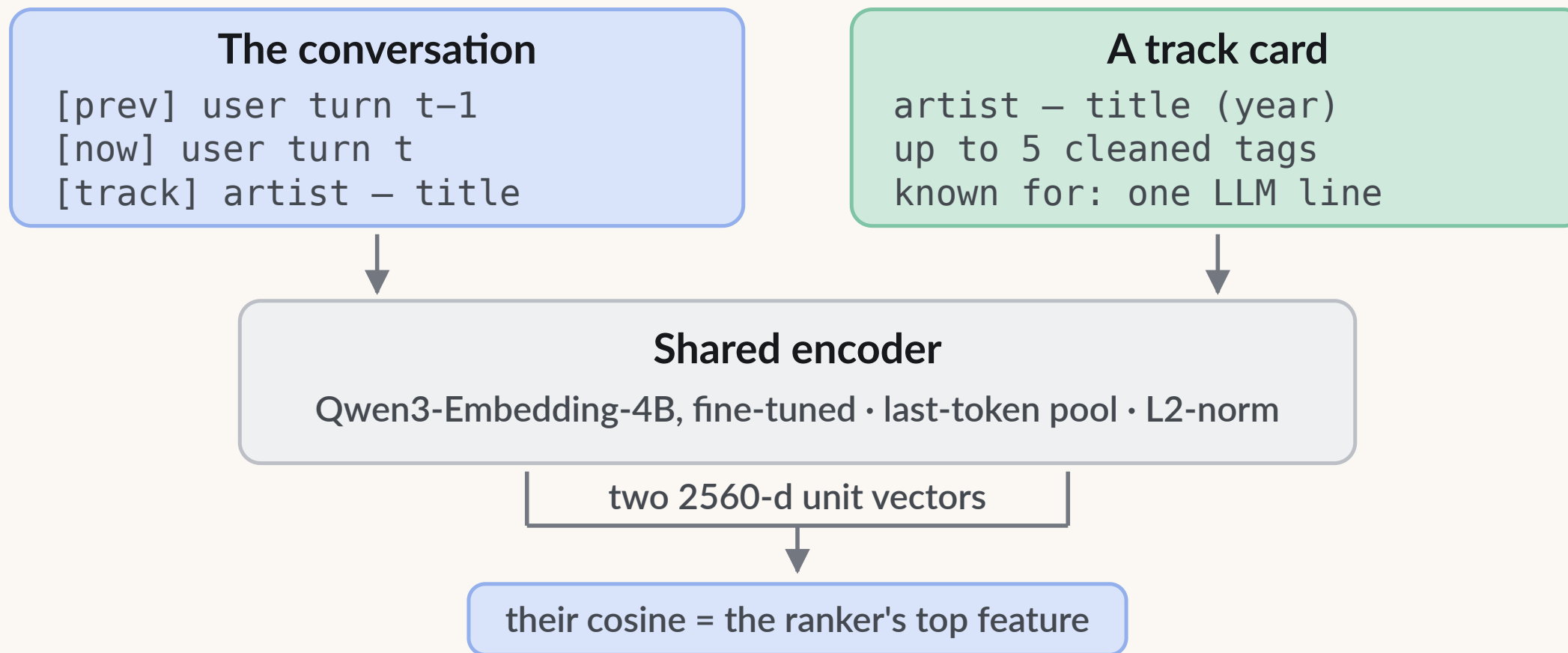
Exact-track pin when a specific track is named; a final artist-constraint check.



The bi-encoder behind that top feature

APPROACH

OPTIONAL





Down-weighting contradicted turns

APPROACH

OPTIONAL

Every turn starts at		1.0
Next turn rejects the artist		× 0.6
Next turn rejects the track		× 0.3
Goal note: did not help		× 0.3
Never below		0.2

WHY

The label is whatever played next. When the **next** turn says it was wrong, that label counts for less in training.

Chosen by hand, never tuned. Goal notes shape these weights but are never a ranker input.



Response generation

APPROACH

OPTIONAL

LISTENER

...it feels a bit too serious. I'm looking for something more uplifting and positive, really something to lift my spirits.

OUR REPLY



“Perfect — ‘Happy Up Here’ by Röyksopp is exactly the kind of uplifting, cheerful electronic track you’re looking for. With its bright synths and hopeful groove, it’s designed to lift your spirits and make you smile.”

A second LLM (**Qwen3-30B**) writes one short reply about the top track. The next slide shows exactly what it was given.



How the reply is written

APPROACH

OPTIONAL

Latest state

Genre	electronic
Mood	uplifting, positive, happy
Avoid	serious

Top track

Track	Röyksopp – “Happy Up Here”
Tags	cheerful, hopeful, experimental, Pop, 00s...



“Write 1–2 concise sentences about only the selected track. Prioritize the latest user request and extracted state over older conversation history. If the track is reasonably aligned, explain the fit with one specific supported reason. If it clearly conflicts with an explicit avoid/new-artist constraint, do not oversell it or blame the system; briefly frame the limitation and the closest supported reason.”

Qwen3-30B-A3B-Instruct

temperature 0

one pass

output: the reply on the previous slide



5 How it scored

the official results

How it scored

RESULTS CORE

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



Where we placed

RESULTS

OPTIONAL

#1 · hallucinated



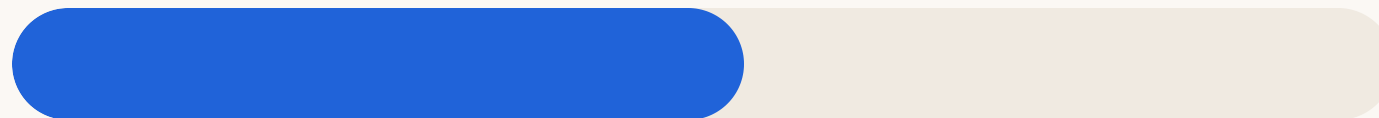
0.689

Median of 41 teams



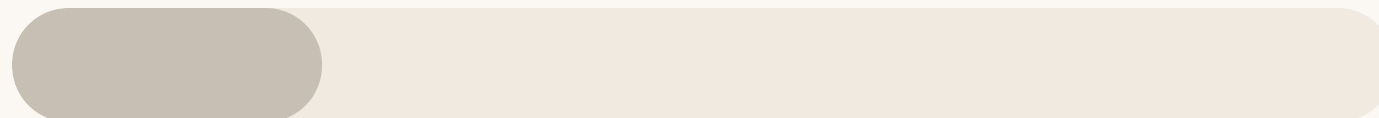
0.474

#29 · npatta01 (us)



0.381

Organisers' BM25 baseline



0.162

29th: well above the baseline, below the median.

Final (Blind-B) composite score. Source: official Music-CRS results table, nlp4musa.github.io/music-crs-challenge/results.html — 41 teams listed; verification in progress.



6 What is not ideal

what went wrong, one finding per slide



What is not ideal

LESSONS

OPTIONAL

We trained and validated the ranker on the **same dev turns**.

THE LISTENER ASKED FOR

We extracted more than we used

*the Kamelot track from **Silverthorn***



OUR TOP PICK

Kamelot — Fallen Star, from **Haven**

We extracted "**Silverthorn**" but never filtered on it

We used no data from outside the catalog

*exactly **126.70 bpm**, in G minor*



the catalog has no tempo or key, and we added no outside source

The reply said "Got it" for a track we did not have

*"**Watercolors**" by Pat Metheny*



Metheny — Alfie, with the reply "Got it — here's 'Alfie'..."

"**Watercolors**" is not in the catalog



Why we trained on the dev set

LESSONS

OPTIONAL



1 paid LLM call per turn
to read each turn into a state



Too costly for 121.6k turns
so no features on the training set



Trained on the 8,000 dev turns
the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.



Extracted, but the ranker did not act on it

WENT WRONG

CORE



*"I gotta break away from **Deltron** and **Del** completely"*

Dead Prez — Hip Hop

state held both artists as avoid

NOT ACTED ON



*"branching out to **other** singer-songwriter artists"*

Ryan Adams — When The Stars Go Blue

state held Ryan Adams as avoid

NOT ACTED ON



*"still focusing on **Juanes**... haven't found the depth I'm after"*

Juanes — Me Enamorado De Ti

state held Juanes as rejected

NOT ACTED ON

We wanted the ranker to use what the state recorded; it served that artist anyway. We think the labels taught it to: they often repeat the just-played artist, so rejections carried only **0.7%** of its decisions.

Where the failures were — all 80 Blind-B turns

ANALYSIS CORE

DIAGNOSED GAP	TURNS	MEANING
Ranking	31	a better, non-violating candidate sat below our top-20
Constraint not applied	8	an extracted filter never reached the pool
State extraction	3	the extractor missed something the user said
Entity resolution	1	a name never resolved to a catalog ID

TOP-1 FLAGGED

17 / 80

the track we led with violated something the state had recorded

HOW THIS WAS COUNTED

43 diagnoses across 32 distinct turns. A **rule-based check** of each turn's system trace — not an LLM, and not a relevance label.

Extraction is **3 of 43** diagnoses. The failure is overwhelmingly downstream of understanding the conversation.



No data from outside the catalog

WHAT IS NOT IDEAL

OPTIONAL

A tempo and a key

“exactly 126.70 bpm, in G minor”



the catalog has no tempo or key field
no branch can match on either

A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”
no film or TV links in the catalog

The state was right both times. The gap is **knowledge, not understanding**.



The reply was never checked

WHAT IS NOT IDEAL

OPTIONAL

The track they asked for is not in the catalog — the reply presents a substitute anyway

1

"I wanted 'Watercolors' by Pat Metheny specifically."

"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

2

"Could you please play Sarah McLachlan's 'Fallen'?"

"You're in the right place — 'Hold On' by Sarah McLachlan is a perfect match..."

NOT FLAGGED

A constraint we did not meet — the reply recommends the track anyway

3

*"...something from a **different artist** this time"*

"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

4

*"...we're still focusing on **Juanes**..."*

"I know you're looking to explore beyond Juanes, but 'Me Enamorado De Ti' is a standout..."

NOT FLAGGED

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).



What the training labels look like

WENT WRONG CORE



“I keep asking for Arcade Fire — that's the only song I want”

label: Arctic Monkeys — This House is a Circus

DOES NOT



“similar chill vibe, but from a different artist”

label: Bonobo — Jets

MOVES



*“what **other** artists fit this darkwave style?”*

label: Tropic of Cancer — The Dull Age

MOVES



*“what **other** orchestral tracks would you recommend?”*

label: Nick Phoenix — Kogan

MOVES

Three of four are annotated **MOVES TOWARD GOAL** — the dataset records them as having helped. Train on this and the model learns to anchor. **Ours does.**

Why: the answer is the simulated recommender's pick from a 16–32-track pool, made without seeing the listener's goal.



The target often ignores the request

ANALYSIS

OPTIONAL

LISTENER · JUST PLAYED BONOBO

...something with a similar chill, electronic vibe, but **from a different artist?**

ANSWER KEY



Bonobo — “Jets”

MOVES TOWARD GOAL

NEXT TURN

“Jets” is cool, but I was hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

The answer key repeats the artist the listener just asked to leave. Our ranker **anchors** the same way — we think it learns it here.

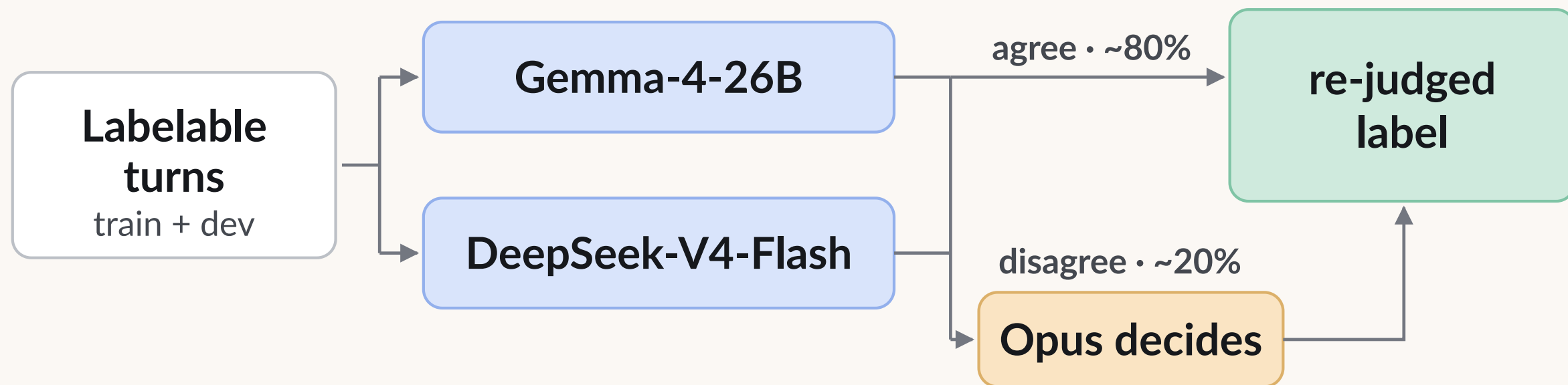


How the LLM relabelling worked

ANALYSIS

OPTIONAL

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



Agreement split measured on the training turns.



What the relabelling found

ANALYSIS

OPTIONAL

113,393

turns re-judged

every labelable train and dev turn

61%

not a good fit

of training turns, the judges would
not settle on the label fitting

17%

keep the rejected artist

18,222 training turns

Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix — so we kept the originals.



Everything is released

THANK YOU

CORE



Paper, poster, slides, code, models, data

npatta01.github.io/music-crs-2026

→ Appendix

TYPE A NUMBER + ENTER · OR G TO SEARCH

- | | | | | | |
|--------------------|----------------------------|--------------------|--|--------------------|--|
| 1 | Where our points came from | 2 | The task | 3 | Glossary |
| 4 | State extraction | 5 | Six request types, six shapes of state | 6 | Routing flags decide which branches fire |
| 7 | Entity resolution | 8 | The eleven branches | 9 | Fusion and fallback |
| 10 | b1 training | 11 | Development-split results | 12 | Failure 3 — the better track was already in the pool |
| 13 | Label-free audit | 14 | Limits of our own analysis | 15 | What we would do differently |
| 16 | Related work | 17 | Reproducing the run | | |

Where our points came from

RESULTS

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat.div} + 0.10 \cdot \text{lex.div} + 0.30 \cdot (\text{judge} - 1)/4$$

OUR BLIND-B DECOMPOSITION

0.50 × nDCG@20	0.2537	0.1269
0.10 × cat. div.	0.0315	0.0032
0.10 × lex. div.	0.7862	0.0786
0.30 × (3.30−1)/4		0.1725
Composite		0.3811

WHERE THE SCORE CAME FROM

The **judge term out-earned nDCG**, and category diversity returned **0.003** of the 0.10 on offer. We optimised almost entirely for relevance and left the diversity terms alone.

EVERY ASSISTANT TURN

- Retrieve **20 tracks** from a shared **47,071-track** catalog
- Write a short reply presenting the top recommendation

PER-TRACK METADATA

Title, artist, album, a median of **17** genre/mood tags, popularity, release date, plus precomputed text, audio, cover-art and collaborative-filtering embeddings.

SPLITS

Train	15,199 sessions (~121.6k turns)
Development	1,000 sessions × 8 turns = 8,000
Blind-A / Blind-B	80 sessions each, final turns

Blind-B additionally withholds the conversation-goal annotations present in training data.

BUILT ON

TalkPlayData 2. The conversations are **LLM-generated**, not observed listening sessions — which matters later.



Glossary

BACKGROUND

Typed state	Structured fields an LLM reads off the conversation each turn: artists to anchor on, rejected tracks, filters, mood.	BM25	A classic keyword-search score over text fields.
Dense embedding	A vector from a neural model; texts with similar meaning land close together.	CLAP	A model that puts text and audio in one space, so a description can find a sound.
SigLIP-2	The same idea for text and images — here, album covers.	Collaborative filtering (BPR)	Vectors learned from who listened to what; BPR is the training method.
Centroid search	Average the vectors of tracks the listener liked, then search near that average.	RRF	Reciprocal-rank fusion: merge ranked lists by summing $1 / (k + \text{rank})$ per track.
LambdaMART · LightGBM	Gradient-boosted decision trees trained to order a list; LightGBM is the library.	Bi-encoder	One model encodes the conversation and the track separately; relevance is the cosine of the two vectors.
LLM judge	An LLM that rates the reply 1–5; the organisers run it.	In-sample vs out-of-fold	Scoring on turns the model trained on, versus on held-out folds. In-sample overstates.



State extraction

APPROACH

THE EXTRACTOR

DeepSeek-V4-Flash (build 0423), schema-constrained, given the dialog so far and the IDs of tracks already played.

DELIBERATELY BLIND

No conversation goals, goal progress, or user profiles reach the extractor. The state is whatever an LLM can read off the conversation.

WHY THIS MODEL

Cheaper extractors under-filled the optional fields — rejections and facets especially — so every final run uses this one. Per-turn extraction is a paid API call, which is what kept us off the training split.

PROMPT VERSION

ConversationStateV1, projected to V0Plus for the downstream contract.



Six request types, six shapes of state

APPROACH

REQUEST TYPE	A REAL TURN	WHAT THE EXTRACTOR PULLED OUT
<code>exact_track</code>	"I wanted <i>Watercolors</i> by Pat Metheny specifically."	track = <i>Watercolors</i> must_use · artist = Pat Metheny must_use
<code>exact_album</code>	"Which song from Peter Doherty's <i>Grace/Wastelands...</i> "	artist + album must_use · facet "best tracks of 2009"
<code>same_artist</code>	"Got another track known for his guitar work?"	artist must_use · prior track do_not_use · facets: guitar work, melodic complexity
<code>new_artist</code>	"...branching out to other singer-songwriter artists."	artist = Ryan Adams & The Cardinals do_not_use · facets: acoustic, introspective
<code>attribute_search</code>	"...a really striking, abstract cover art, bold colors and shapes."	three visual facets · routing flag image_or_visual_search
<code>hidden_target</code>	"A Lil Wayne song from an experimental phase."	artist must_use · facet "experimental" · flag hidden_target_search

The extractor read all six of these correctly. Across the whole Blind-B run only **3 of 80 turns** were diagnosed as a state-extraction miss — reading the conversation was almost never the problem.



Routing flags decide which branches fire

APPROACH

FLAG	OPENS
exact_entity_probe	catalog lookups — discography, album
lyric_search	the Qwen3 lyrics view
feature_articulation	CLAP text→audio on the sonic description
image_or_visual_search	the gated SigLIP-2 cover-art branch
hidden_target_search	widens the pool; suppresses exact pins

A REAL VISUAL REQUEST

“I’m looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets, `image_or_visual_search = true`

AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.



Grounding names to the catalog

APPROACH

NAMES → IDS

Fuzzy string matching of artist and track names against the catalog name lists, keeping matches above **80/100**. Downstream stages only ever see identifiers.

FREE TEXT → TAGS

- 1 Exact match against the tag vocabulary
- 2 Alias match
- 3 Substring match
- 4 Embedding nearest neighbour, cosine ≥ 0.6 , top 3

Used by the BM25 branch. Outside it, tag matching falls back to exact strings — part of why so little of the state became executable.

The eleven retrieval branches

APPROACH

BRANCH	SPACE	QUERY
BM25 (fielded)	lexical	state text + resolved tags
Dense text × 3	Qwen3 views	state-built strings
CLAP text→audio	audio	sonic description
Anchor centroids × 3	SigLIP-2/CLAP/CF	liked-track centroids †
User-CF centroid	CF-BPR	the listener's own vector
Lookups × 2	catalog	discography; era popularity

GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.



Weighted RRF — baseline and fallback

APPROACH

FUSION

Weighted reciprocal-rank fusion, $k = 60$, top-1000 across the branch pools.

POST-FUSION ADJUSTMENTS

- Drop rejected artists and tracks
- Drop tracks already played
- Promote / demote on exact tag membership
- Year-range decay

AS A BASELINE

0.1492

devset nDCG@20, no training

STILL THE FALLBACK

If the learned ranker is unavailable, this order ships. It is the only part of the ranking stack with no trained component.



How b1 was trained

APPROACH

$$L = -(1/B) \sum_i \log \left[\exp(v_{q_i}^T v_{d_{i+}} / \tau) / \sum_{d \text{ in } D_i} \exp(v_{q_i}^T v_d / \tau) \right] \quad \tau = 1/20$$

POSITIVES

53,885

(turn, track) pairs whose *next* turn's goal-progress says the recommendation moved the session forward

NEGATIVES

4 + batch

four mined hard negatives per positive, plus every in-batch document

RETRIEVAL VS RANKING

As a twelfth *retrieval* branch it added little to the union's coverage and was dropped. As a *ranking* feature it dominates the fit — out-of-fold nDCG@20 0.1970 → 0.2032.

Not isolated: which parts of the rendering matter, and whether the effect carries to the blind splits.



Development split — what we steered on

RESULTS

SYSTEM (HOW MEASURED)	NDCG@20
Weighted RRF fusion (no training)	0.1492
LambdaMART, out-of-fold CV	0.1970
+ b1_cos (deployed feature set)	0.2032
Historical v10 LambdaMART, in-sample	0.3844
Official Blind-B	0.2537

DO NOT OVER-READ THE FIRST GAP

The fusion and learned-ranker configurations used **different candidate-pool depths**, so 0.149 vs 0.197–0.203 does not isolate the effect of learned re-ranking.

THE IN-SAMPLE ROW

A historical goal-aware model without b1_cos.
Not directly comparable — shown because it is the number we were selecting on.



Failure 3 — the better track was already in our pool

ANALYSIS

WHAT THEY ASKED FOR	WE SERVED AT RANK 1	ALREADY IN OUR POOL	POOL RANK
nu-jazz, downtempo, female vocals — <i>not</i> Parov Stelar	Parov Stelar — Booty Swing	St Germain — What You Think About...	34
new underground hip-hop, not Deltron	Dead Prez — Hip Hop	Souls Of Mischief — Make Your Mind Up	31
hardcore, not classic hip-hop	Dead Prez — Hip Hop	Papa Roach — Getting Away With Murder	49
that carol, but not Beegie Adair again	Beegie Adair — The Christmas Song	Frank Sinatra — Have Yourself A Merry Little Christmas	63

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.



Auditing the submitted run

ANALYSIS

SINGLE-JUDGE QUALITY RATING

78%

of Blind-B turns rated weak-or-bad by one DeepSeek-V4-Flash judge

READ AS

Diagnostic, not ground truth. Blind-B exposes no relevance labels, so only the validation failure is actually *measured*.

DETERMINISTIC TRACE AUDIT — 43 DIAGNOSES OVER 32 TURNS

Heuristic alternative below top-20	31
Under-applied constraint	8
State-extraction miss	3
Entity-resolution miss	1

Categories overlap. The alternative was chosen by positive-term metadata matches plus popularity — an audit heuristic, not a relevance judgment.



Limits of our own analysis

ANALYSIS

- The relabeling was **never checked against human judgments**.
- The Blind-B quality audit used a **single** LLM judge.
- The trace audit was **heuristic**, not a relevance judgment.
- The conversations are **LLM-generated**, not observed listening sessions.
- Neither blind split exposes relevance labels — blind-set failure rates and relabeling rates are **diagnostic, not ground truth**.
- Development-split and leaderboard numbers are **separate measurements**.



What we would do differently

CONCLUSION

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

→ Related work — four established lines

A1

CONVERSATIONAL STATE TRACKING

CRS systems commonly track dialog state explicitly and condition retrieval on it. We follow that, but compile the state with a **prompted LLM** instead of a trained tracker.

MODALITY-SPECIFIC MUSIC ENCODERS

Contrastive language-audio pretraining (CLAP), vision-language over cover art (SigLIP-2), collaborative filtering (BPR), general text embeddings (Qwen3).

RETRIEVE-THEN-RERANK AT CATALOG SCALE

Pool multiple sources with reciprocal-rank fusion; rank with LambdaMART.

OUR CLAIM

What we report is **how that combination behaved** — not a new technique.

→ Reproducing the run

A2

FROZEN REPLAY – NO CREDENTIALS

Source the offline bundle's env and every LLM call is served from a curated cache. A miss hard-fails rather than silently spending money.

```
source .repro/scripts/activate_repro_env.sh
```

LIVE RE-RUN

```
python run_experiment.py --backend modal --tid state_ranker_v10_lgbm_blindset_B --eval_dataset blindset_B
```

ARTIFACTS

Models and data: [hf.co/datasets/Npatta01/music-crs-repro-2026](https://huggingface.co/datasets/Npatta01/music-crs-repro-2026)